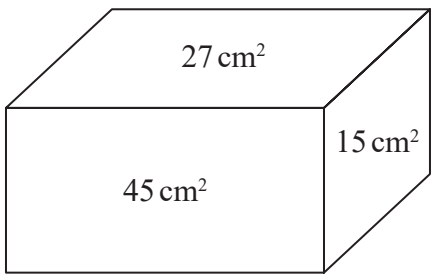
8 The diagram shows a solid metal cuboid.

The areas of three of the faces are marked on the diagram.
The lengths, in cm, of the edges of the cuboid are whole numbers.



The metal cuboid is melted and made into cubes.
Each of the cubes has sides of length 2.5 cm.

Work out the greatest number of these cubes that can be made.

$45 = 5 \times 9$, $27 = 3 \times 9$, $15 = 3 \times 5$

So the three edge lengths are 3, 5 and 9 cm.

1 mark
Find three dimensions consistent with all three face areas.

Cuboid volume $= 3 \times 5 \times 9 = 135\ cm^3$

1 mark
Multiply the three dimensions.

One cube: $2.5^3 = 15.625\ cm^3$

1 mark
Find the volume of one cube.

$135 \div 15.625 = 8.64$

1 mark
Compare the two volumes.

Maximum = 8 complete cubes

1 mark
Round down to the number of whole cubes.

(Total for Question 8 is 5 marks)